

MARKET BRAIN GRID

Product Requirements Document

Indonesia equities + crypto | Hermes multi-agent runtime | Local-first with Supabase/Firebase-ready cloud deployment

Version 1.0 - Build Specification Draft

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Core product principle: the LLM proposes and explains; deterministic services validate data, calculate risk, run backtests, and gate execution.

Dimension	Decision
Primary role	AI research desk + portfolio decision support + backtesting + optional paper/live execution in later phases
Initial markets	Indonesia listed equities (IDX) and spot crypto; derivatives only after explicit phase-gated support
Primary interface	Hermes Agent via CLI/chat/messaging; web dashboard for monitoring and artifacts
Default mode	Research / paper mode; live trading disabled until safety gates are satisfied
Data principle	Every market fact must have timestamp + source + instrument identity + unit/caliber metadata
Deployment	Local laptop first; Supabase recommended cloud data/control plane; Firebase optional for UI/event plumbing
Decision output	Evidence -> analyst reports -> adversarial debate -> risk committee -> portfolio decision -> plan

Document purpose

This PRD translates the requested concept into a buildable architecture for a Hermes-driven trading system. It combines patterns observed in AutoHedge, TradingAgents / arXiv:2412.20138, Vibe-Trading, Moss trade-bot skills, and the supplied ChatGPT Astra research workflow, while adapting the design to Indonesia equities and crypto.

This is a software product specification, not investment advice. It explicitly separates research output from execution authority and requires deterministic safety controls outside the language model.

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1. Executive summary

is a research-first multi-agent trading system orchestrated by Hermes. Its job is to continuously turn market data into auditable analysis, ranked opportunities, trade plans, backtests, and portfolio/risk assessments. It should support both Indonesian equities and crypto without pretending they follow the same market microstructure.

The architecture deliberately combines the best elements found in the researched systems: TradingAgents-style specialist analysts and bull/bear debate; AutoHedge-style director/quant/risk/executive separation and structured output; Vibe-Trading-style market-tool abstraction, backtests, persistent memory, shadow account, auditability, and data-quality guards; Moss-style parameterized strategies, backtest/live parity, and bounded evolution; and the supplied Astra workflow's insistence on source-grounded facts, explicit risk math, invalidation, confidence, and human approval.

North-star outcome

Every trading day, should produce a compact decision packet for IDX and crypto that answers: what changed, what matters, which instruments deserve research, what the evidence says, what could invalidate the thesis, what the portfolio risk is, and what exact conditions would be required to act. The system must be able to say NO TRADE when evidence or data quality is insufficient.

Recommended build stance

Area	Recommendation	Why
Runtime	Hermes Agent as human-facing orchestrator + skill loader	Matches requested operating model; skills are designed for on-demand workflows and external tools.
Core backend	Python services with typed schemas	Best fit for quant libraries, broker/exchange adapters, backtesting and data engineering.
DB/control plane	Supabase/Postgres	Strong fit for relational market metadata, runs, audit, RLS, scheduled jobs, realtime and vectors.
Frontend	Next.js / React dashboard	Fast operator console for reports, positions, backtests and approvals.
Messaging	Hermes-native channels first; Telegram/Discord/web later	Keeps user workflow close to agent and expands only after core stability.
LLMs	Provider-agnostic model router	Supports cost/quality tiers and reduces lock-in.
Execution	Separate deterministic execution service	Prevents prompt-level hallucination from becoming an order.

Important architectural choice: do not embed exchange secrets or direct order methods inside analyst/researcher skills. Only a hardened execution gateway may create an order request, and every request must pass deterministic pre-trade checks.

2. Product vision, scope and non-

2.1 Vision

Build a personal institutional-style trading research desk that is always available through Hermes, remembers research context, can replay exactly how a recommendation was produced, and can move from idea -> data -> backtest -> paper trade -> guarded live execution without rewriting the research stack.

2.2 Scope

Capability	MVP	Future
Daily IDX market brief	YES	Intraday streaming
Daily crypto brief	YES	24/7 alerting + regime monitoring
Candidate scanner	YES	Adaptive universe discovery
Technical/fundamental/news/sentiment research	YES	Specialized alternative datasets
Bull/bear debate	YES	Multi-round committee optimization
Risk and position sizing	YES	Portfolio optimizer / risk parity
Backtest engine	YES	Walk-forward, ensembles, meta-labeling
Paper trading	YES	Exchange-specific paper/sandbox where available
Live execution	GATED	Multi-venue smart routing
Trade journal/shadow account	YES	Behavior analytics + coaching
Strategy evolution	P1	Automated bounded tuning
Mobile/voice	P2	Personalized alerts

2.3 Non-goals

- No guarantee of profits or prediction accuracy.
- No use of LLM output as the sole source of market prices, balances, fills, or order state.
- No direct broker/exchange access from the research agents.
- No unbounded autonomous self-modification of live trading parameters.
- No mixing of IDX and crypto market rules in one generic execution policy.
- No backtest reports that omit fees, slippage, funding/financing where applicable, corporate-action treatment, or data-quality status.

2.4 Operating modes

Mode	Behavior	Execution
RESEARCH	Collect/analyze/report; no simulated positions required	NONE
PAPER	Simulated portfolio; real-time or delayed market data	SIMULATED
SHADOW	Observe live account and produce what would have done	NONE
GUARDED LIVE	Live orders only after explicit approval and risk gateway checks	ALLOWLISTED
EMERGENCY STOP	Cancel/reject new risk-taking orders; reconcile state	NONE / CLOSE ONLY if separately authorized

3. Reference-project analysis

3.1 Supplied ChatGPT Astra PDF

The supplied guide is a strong baseline for research discipline. It defines a five-step sequence - SCAN, ANALYZE, RESEARCH, RISK, PLAN - where each stage feeds the next and the final plan is explicitly awaiting human review. It also requires dated/source-backed inputs, a FACTS versus OPINION split, explicit invalidation, position-size math, and a minimum 2:1 risk/reward threshold. The document explicitly states that its "bot" is a repeatable research system rather than an auto-trader and that nothing is executed.

should retain these strengths, but move them into machine-enforced workflow stages, typed schemas, persistent evidence records, deterministic calculations, portfolio-aware risk, and asset-specific market adapters. In particular, should not rely on a user manually pasting each previous step.

3.2 TradingAgents / arXiv:2412.20138

TradingAgents models a trading firm with specialist fundamental, sentiment, news and technical analysts, bullish and bearish researchers, a trader, a risk team and portfolio manager. The current project has expanded its configuration surface and multi-provider model support. The central lesson for is role separation plus adversarial debate and explicit final portfolio/risk review.

improvement: split analysts by market context (IDX equity vs crypto), separate research truth from decision logic, and add a deterministic risk engine and data-quality layer that can veto agent output before it reaches a portfolio decision.

3.3 AutoHedge

AutoHedge emphasizes a Director -> Quant -> Risk Manager -> Execution Agent pipeline, real-time market data, structured JSON output, detailed logging, and modular venue support. It currently emphasizes autonomous Solana trading and has a narrower venue footprint than requires.

improvement: treat execution as a downstream gateway rather than a normal LLM role; expand venue adapters; add full audit events, paper/shadow modes, deterministic reconciliation and stronger market-data provenance.

3.4 Vibe-Trading

Vibe-Trading is the strongest reference for a complete research workspace. Its current scope includes market-data routing, multiple backtest engines, persistent research memory, shadow-account behavior diagnostics, multi-agent research teams, MCP interfaces, and strong data-quality defenses. Its recent change log also demonstrates how subtle failures can invalidate trading research: unit mismatches, wrong instrument identity, lookahead from announcement timestamps, silent data-source substitution, price-caliber mismatches, execution-cap errors, and reports incorrectly passing when evidence was missing.

should adopt this mindset early. Data quality is not a helper utility; it is a first-class gate in the trading pipeline.

3.5 Moss trade-bot skills

Moss adds an appealing strategy factory: natural-language descriptions become parameterized strategies, backtests calculate risk/return metrics, and a bounded evolution loop micro-adjusts tactical parameters while keeping core personality parameters locked. The current skill is designed for simulated Hyperliquid perpetuals and supports more than 30 tunable parameters, safety guards, and bit-exact alignment between backtest and live strategy logic.

improvement: use the same backtest/live parity p
equities and crypto spot first, with market-rule adapters so the same strategy cannot accidentally inherit the
wrong T+1, leverage, contract, lot, session, or funding rules.

3.6 Hermes Agent

Hermes provides an on-demand Skills system using SKILL.md documents, progressive disclosure, external skill directories, and config injection. Its documentation explicitly recommends using a Skill for repeatable workflows backed by existing tools and a Tool for capabilities requiring custom processing, credentials, or strict binary/streaming integration. That maps cleanly to : research procedures should be Hermes skills; data/exchange/broker operations should be deterministic services or tools behind a narrow interface.

4. product requirements

4.1 Functional requirements

ID	Requirement	Acceptance summary	Priority
FR-001	Daily market brief	Produce one IDX brief and one crypto brief with market regime, movers, catalysts, risk events and top research candidates.	P0
FR-002	Candidate ranking	Rank candidates by evidence quality and setup clarity, not expected return alone.	P0
FR-003	Evidence ledger	Store each fact with source, timestamp, instrument ID, unit, and provenance.	P0
FR-004	Analyst fan-out	Run technical, fundamental/on-chain, news, sentiment, macro and market-structure analysis in parallel.	P0
FR-005	Adversarial review	Run Bull and Bear researchers against the same evidence packet.	P0
FR-006	Risk committee	Evaluate liquidity, concentration, volatility, stop distance, scenario risk and portfolio impact.	P0
FR-007	Trade plan	Create entry/invalidations/targets/position size/expected holding period and verifiable assumptions.	P0
FR-008	Backtest	Run reproducible historical tests with deterministic data snapshot and strategy version.	P0
FR-009	Walk-forward	Support train/test or parameter-selection windows separated in time.	P1
FR-010	Paper/shadow	Simulate or shadow recommended actions against live data.	P0
FR-011	Order gateway	Route only approved, validated orders; reject malformed or unsafe requests.	P1
FR-012	Kill switch	One-click or one-command halt for new orders across all venues.	P0
FR-013	Journal	Persist decisions, orders, fills, outcomes and post-trade review.	P0
FR-014	Memory	Recall similar past setups and prior outcomes without contaminating current facts.	P1
FR-015	Strategy lab	Parameter sweeps, fingerprints, validation, and comparison of strategy versions.	P1
FR-016	Alerts	Notify on thesis invalidation, risk breach, data outage, fill mismatch and system errors.	P0

4.2 Non-functional requirements

NFR	Target
Reproducibility	Same run ID + same data snapshot + same strategy version must reproduce deterministic calculations exactly.
Auditability	Any recommendation must resolve to agent outputs, evidence IDs, model/version metadata and risk calculations.
Safety	No path from natural language directly to exchange order without policy gateway.
Data freshness	Each source declares expected freshness; stale data must lower confidence or block action.
Availability	Research mode can degrade gracefully when one source fails; execution mode fails closed.
Extensibility	New venue, data vendor, strategy or model added through adapter interface, not core rewrites.
Cost control	Use cheap/fast models for parsing and routine summaries; deep model only for synthesis/debate where justified.
Privacy	Secrets isolated from prompts; logs redact credentials and sensitive account details.

5. Hermes agent roles and skills

Hermes is the user-facing operator and workflow runner. should expose a small number of stable Skills, each backed by deterministic Python commands or API calls. The goal is not to create dozens of LLM personalities; it is to make each reasoning role accountable for one artifact and one schema.

5.1 Core agent roster

Role	Agent	Primary job	Hard boundary	Main skill/tool
ORCHESTRATOR	Director	Decides which workflow to run, creates run IDs, routes tasks, checks prerequisites, and stops when a hard gate fails.	No market facts. Never sizes positions.	run_daily_desk, run_asset_research
DATA STEWARD	Data Guardian	Fetches market/fundamental/news datasets, validates identity/units/time/coverage, records provenance and data quality.	May not form a trade thesis.	fetch_market_data, validate_dataset
IDX TECHNICAL	IDX Chart Analyst	Technical analysis for Indonesian equities using venue-specific calendars, price limits, liquidity and corporate-action metadata.	No order creation.	analyze_idx_technical
CRYPTO TECHNICAL	Crypto Market Analyst	Technical and market-structure analysis across selected exchanges/pairs, including volatility, funding where applicable, liquidity and regime.	No order creation.	analyze_crypto_technical
FUNDAMENTAL / ON-CHAIN	Research Analyst	Equity financials and crypto fundamentals/on-chain/network metrics.	Facts must be sourced and dated.	research_fundamentals, research_onchain
NEWS & SENTIMENT	Catalyst Analyst	News, announcements, social/sentiment and event risk; labels rumor vs confirmed.	Never infer a price from a headline alone.	research_news_sentiment
MACRO	Macro Analyst	Rates, FX, commodities, policy and global risk factors relevant to Indonesia and crypto.	No instrument-specific execution.	research_macro
QUANT	Quant Researcher	Transforms analyst facts into quantitative signals, factors, statistics and regime scores.	Must distinguish computed values from sourced facts.	compute_factors, score_candidates
BULL RESEARCHER	Bull Case	Builds strongest upside case and identifies conditions required for it to hold.	Cannot approve its own thesis.	debate_bull
BEAR RESEARCHER	Bear Case	Builds strongest downside case, failure modes and alternative explanations.	Cannot approve its own thesis.	debate_bear
RESEARCH MANAGER	Debate Chair	Runs structured debate and synthesizes disagreements into a Research Verdict.	Cannot bypass risk gate.	run_debate
RISK COMMITTEE	Risk Guard	Deterministic risk service + optional LLM committee; evaluates position sizing, exposure, liquidity and scenarios.	Hard veto authority.	risk_check, size_position
PORTFOLIO MANAGER	Portfolio Decision	Combines research and	Never directly calls	make_portfolio_decision

		risk output WATCH / PAPER / PROPOSE LIVE decision.	exchange APIs.	
EXECUTION GATEWAY	Order Sentinel	Deterministic service only; validates and routes orders after approval.	No LLM reasoning inside the critical path.	submit_order, cancel_order, reconcile
POST-TRADE REVIEWER	Journal Analyst	Explains realized vs expected outcome and updates strategy diagnostics.	Cannot alter live parameters directly.	review_trade, update_journal
STRATEGY EVOLVER	Research Lab Agent	Runs bounded parameter evolution and rejects changes without out-of- sample improvement.	Cannot mutate production live config automatically.	evolve_strategy

5.2 Hermes skill layout

```

~/hermes/skills/finance/      -trading/
SKILL.md                      # top-level workflow + routing rules
references/
  market-schema.md           # canonical market/evidence schema
  risk-policy.md              # policy version and hard limits
  idx-rules.md                # venue rules/config source
  crypto-rules.md             # exchange/asset rules
  report-templates.md         # output structures
scripts/
  .py                         # CLI facade
  run_daily.py
  run_backtest.py
  healthcheck.py
templates/
  daily_brief.json
  trade_plan.json
  backtest_report.json

```

5.3 Skill activation rules

- Use fast/small models for extraction, classification and formatting; use the strongest reasoning model for debate and synthesis.
- Every skill starts by checking data readiness and instrument identity.
- Any missing prerequisite returns a typed BLOCKED status, not a fabricated answer.
- All tool calls are logged with run ID, agent role, latency, source, and result status.
- Live execution skills are unavailable unless system mode is GUARDED LIVE and policy gates pass.

6. Workflow and decision lifecycl

6.1 Daily desk - master workflow

SCHEDULE -> DATA HEALTH -> SCAN -> ANALYZE -> RESEARCH -> DEBATE -> RISK -> PORTFOLIO DECISION -> PLAN
-> ALERT -> PAPER/SHADOW -> JOURNAL

6.2 version of the supplied five-step workflow

Stage	Original idea retained	upgrade
SCAN	Find 3-5 candidates; rank clarity, not return	Multi-market screener + data-quality score + liquidity filters + portfolio context
ANALYZE	Trend, momentum, support/resistance, price action	Deterministic technical features + chart narrative; separate visible facts from interpretation
RESEARCH	Financials, earnings, news, catalysts	Source graph + event-time knowledge + equity vs crypto research templates
RISK	Stop, size, R:R, invalidation	Portfolio-aware risk budget + stress tests + correlation + liquidity + concentration + hard policy veto
PLAN	One structured plan awaiting review	Proposal object with evidence IDs, expiry time, scenario tree, execution constraints and status machine

6.3 Candidate state machine

DISCOVERED -> DATA_VALIDATED -> ANALYZED -> DEBATED -> RISK_CHECKED -> PROPOSED -> PAPER
-> APPROVED -> EXECUTED -> MONITORED -> CLOSED -> REVIEWED
Any stage -> BLOCKED / REJECTED / EXPIRED
Any live position -> RISK_BREACH -> EMERGENCY_ACTION -> RECONCILED

6.4 Recommendation statuses

Status	Meaning
NO SETUP	No sufficiently clear opportunity.
WATCH	Interesting but missing confirmation or price condition.
PAPER CANDIDATE	Passes research + deterministic risk in simulation.
LIVE PROPOSAL	Ready to be reviewed under live policy; not an order.
APPROVED	User explicitly approves a specific proposal ID and constraints.
REJECTED	Proposal failed policy, evidence or review.
EXPIRED	Conditions changed or data is too old; must be regenerated.

7. Market-specific intelligence

7.1 Indonesia equities (IDX)

must treat Indonesian equities as a first-class market adapter rather than forcing them through crypto assumptions. The adapter should own trading sessions, trading calendar, instrument identifiers, lot/board rules, corporate actions, settlement metadata, price-limit rules, liquidity classification, short-selling availability, and broker-specific execution constraints. Production market data should come from an authoritative or appropriately licensed provider; public/secondary APIs can be used for research or cross-checking when their terms and quality permit.

IDX research dimensions

- Price/volume trend, relative strength, volatility and support/resistance.
- Financial statement trend, margins, cash/debt, valuation, earnings revisions and corporate actions.
- Foreign/domestic flow proxies where reliable and permitted.
- Commodity and FX sensitivity for relevant sectors.
- Government/regulatory/event catalysts.
- Liquidity, gap risk, price-band behavior and tradability for the proposed position size.

7.2 Crypto

Crypto is a 24/7 market with exchange fragmentation. therefore needs exchange-aware market data, pair normalization, quote-currency normalization, order-book/liquidity metrics, funding/open interest only for derivatives, and venue-specific fee/slippage assumptions. Spot and derivatives must be separate risk domains.

Crypto research dimensions

- Price/volume/momentum and volatility regimes.
- Order-book depth, spread, imbalance and liquidity decay.
- Exchange flows, on-chain activity and token-specific events when data is available.
- Funding, basis, open interest and liquidation context for derivatives.
- BTC/ETH dominance or benchmark-relative behavior.
- 24/7 event risk, exchange outage risk and stablecoin/liquidity risk.

7.3 Cross-market macro layer

A macro layer should contextualize, not dictate, individual trades. Useful shared factors include USD/IDR, global rates, commodity prices relevant to Indonesia, global equity risk sentiment, Bitcoin risk regime, and major central-bank or domestic policy events. Macro scores must remain explicitly separate from instrument-specific evidence.

8. Data architecture and provenance

8.1 Canonical data flow

Providers -> Raw Capture -> Normalizer -> Quality Gate -> Feature Store -> Evidence Ledger -> Agent Context -> Decision Artifacts

8.2 Provider abstraction

Adapter	Responsibilities	Examples / direction
IDX Market Data	OHLCV, corporate actions, calendar, instrument metadata	Licensed IDX/broker/provider feed; secondary cross-checks
Crypto Market Data	Ticker, OHLCV, trades, order book, exchange metadata	Native exchange APIs + CCXT adapter where appropriate
Fundamental Data	Financial statements and corporate events	Provider adapters; company filings
News	Timestamped news/event feed	Licensed news APIs, RSS/web extraction where permitted
Sentiment	Social/public chatter	Careful source labeling; never treat as verified fact
Macro	Rates, FX, commodities, economic calendar	Authoritative economic/market sources

8.3 Evidence object

```
{
  "evidence_id": "ev...",
  "instrument_id": "IDX:BBCA",
  "kind": "price|fundamental|news|macro|onchain|derived",
  "observed_at": "2026-09-08T09:00:00+07:00",
  "published_at": "...",
  "source": "provider_name",
  "value": 123.45,
  "unit": "IDR",
  "price_caliber": "raw|adjusted|unknown",
  "payload_hash": "sha256...",
  "quality": "PASS|WARN|FAIL"
}
```

8.4 Mandatory data-quality gates

Gate	Reject / warn condition
Identity	Ambiguous ticker, wrong market or quote currency.
Freshness	Data older than policy threshold for the requested timeframe.
Completeness	Missing bars, missing corporate-action records, missing critical fields.
Units	Volume or contract size cannot be identified with confidence.
Caliber	Adjusted/raw price treatment is mixed without explicit reconciliation.
Time	Published/announcement time unavailable when event timing matters.
Source integrity	Provider returns HTML/error/empty payload where numeric data is expected.
Cross-source discrepancy	Primary vs secondary values differ beyond configured tolerance.
Lookahead	Evidence was not public by the simulated decision timestamp.

8.5 Data snapshot and fingerprinting

Every backtest and high-impact recommendation should be recorded. This is essential for later debugging and for proving that a changed provider did not silently change a backtest result.

9. Backtesting and research lab

9.1 Design principles

- Backtests must use the same strategy interface as paper/live execution.
- Signal time and fill time are distinct. A signal generated from bar T cannot fill at a price that was unavailable until after the decision point.
- Fees, slippage, spread, market impact and funding/financing must be configurable and reported.
- Corporate-action and price-adjustment policy must be explicit for equities.
- Execution constraints must be venue-specific: sessions, lot size, leverage, short availability, position rules, etc.
- Every result must have a data-quality verdict and a leakage check.

9.2 Backtest types

Type	Use
Single-strategy	Validate one strategy on one instrument or universe.
Portfolio backtest	Test multi-asset allocation with shared capital and risk budget.
Walk-forward	Tune only on historical training window; score on later unseen window.
Monte Carlo / bootstrap	Estimate path sensitivity and downside variability.
Stress test	Shock spread, slippage, volatility, gaps and correlated drawdowns.
Event replay	Re-run around specific news/catalyst periods.
Shadow comparison	Compare proposal against naive baseline or benchmark.
Evolution backtest	Bounded parameter search with fixed personality/risk constraints.

9.3 Minimum metric set

Category	Metrics
Return	Cumulative return, CAGR/annualized return where meaningful, benchmark-relative return
Risk	Max drawdown, downside deviation, VaR/ES (where statistically defensible), volatility
Trading	Win rate, payoff ratio, expectancy, turnover, average hold time
Efficiency	Sharpe, Sortino, Calmar; always disclose methodology
Execution	Average slippage, spread cost, fee cost, fill ratio, rejected orders
Robustness	Out-of-sample score, parameter sensitivity, regime breakdown, bootstrap confidence

9.4 Anti-leakage checklist

- Use `published_at` rather than `period_end` for news/fundamental availability when applicable.
- Never fill at the same close used to calculate the signal unless the strategy explicitly models auction/close execution.
- Separate universe selection from future constituents to avoid survivorship bias.
- Use historically valid corporate-action data and do not mix adjusted prices with raw execution assumptions.
- Prevent future funding rates, future index membership, or post-event labels from entering prior bars.
- Use the same lot/contract/fee rules that would have been known at the time.

9.5 Strategy object

```
{
  "strategy_id": "          .breakout.v1",
  "market": "IDX|CRYPTO",
```

```
"universe": {...},  
"timeframe": "1d|4h|1h|15m",  
"signals": [...],  
"risk_policy_id": "rp_default_v1",  
"execution_model_id": "exec_idx_v1",  
"parameters": {...},  
"version": "1.0.0"  
}
```

10. Risk, safety and execution co

Safety rule: LLM outputs are untrusted inputs. The risk engine, order gateway, policy engine and reconciliation service are the system of record for money-moving state.

10.1 Risk hierarchy

Portfolio limits
 -> Strategy limits
 -> Instrument limits
 -> Position sizing
 -> Order limits
 -> Exchange/broker constraints

10.2 Minimum hard limits

Control	Default posture	Configurable?
Max risk per idea	Small fraction of portfolio risk budget; example starting point 0.5-1% for paper research	YES
Max portfolio drawdown stop	Disable new risk-taking after threshold	YES
Max daily loss	Hard halt on new orders when breached	YES
Max position notional	Per instrument and portfolio	YES
Max concentration	Per asset, sector/theme and venue	YES
Max leverage	1x by default until derivatives are separately approved	YES
Liquidity cap	Position/order size limited by estimated executable liquidity	YES
Stale-data block	No live order when required price/account data is stale	YES
Kill switch	Reject new orders and surface emergency state	NO (mandatory)
Manual approval	Required before first live deployment and configurable after	YES, but cannot be removed from high-risk operations

10.3 Risk calculations

```

risk_budget = portfolio_equity * max_risk_pct
risk_per_unit = abs(entry_price - stop_price)
raw_qty = risk_budget / risk_per_unit
final_qty = floor_to_venue_constraints(raw_qty)
notional = final_qty * worst_case_entry_price
portfolio_after = current_exposure + proposed_exposure

```

The implementation must use the worse of quote price and limit price when sizing a limit order, and it must reject ambiguous or unparseable prices instead of silently defaulting.

10.4 Scenario tests before live proposal

- Normal case: expected spread and slippage.
- Worse fill: spread + configured slippage multiplier.
- Gap case: stop filled materially beyond stop price.
- Liquidity case: only a fraction of desired quantity is fillable.
- Correlated shock: portfolio correlation rises while the trade moves against it.
- Venue outage: order submission succeeds ambiguously; system must reconcile before retrying.

10.5 Execution state machine

```

PROPOSED -> APPROVED -> PRECHECK -> SENT -> ACK/UNKNOWN -> PARTIAL -> FILLED
          |                               |

```

REJECT

RECONCI

|

CANCEL / RETRY only if idempotency rules permit

10.6 Emergency behavior

Emergency stop is a system state, not a chat instruction. When active, all new risk-taking orders are rejected. The system surfaces open positions, stale account state, exchange health and outstanding order uncertainty. Any close-only or hedge action must have a separate deterministic emergency policy.

11. Storage model and API contra

11.1 Core tables

Table	Key fields
users	id, auth_provider, preferences, timezone
portfolios	id, user_id, base_currency, mode, risk_policy_id
venues	id, type, market, status, adapter_version
instruments	id, venue, symbol, asset_class, quote_currency, rule_config_id
market_bars	instrument_id, timeframe, ts, OHLCV, source_id, quality
evidence	evidence_id, instrument_id, source, timestamps, payload_hash, quality
research_runs	run_id, workflow, status, started_at, model_versions, code_version
agent_artifacts	run_id, agent_role, artifact_type, schema_version, payload
recommendations	recommendation_id, run_id, status, thesis, invalidations, expiry
risk_checks	recommendation_id, policy_version, checks, pass/fail, calculations
strategies	strategy_id, version, market, parameters, execution_model_id
backtest_runs	run_id, strategy_id, data_snapshot, metrics, leakage_verdict
paper_orders	order_id, portfolio_id, recommendation_id, status, fills
live_orders	order_id, venue_order_id, status, reconciliation_state
positions	portfolio_id, instrument_id, qty, avg_price, unrealized_pnl, risk
journal_entries	trade_id, thesis, outcome, deviation, lesson
alerts	alert_id, severity, channel, status, acknowledged_at
audit_events	event_id, actor, action, object_type, object_id, payload_hash

11.2 Recommendation contract

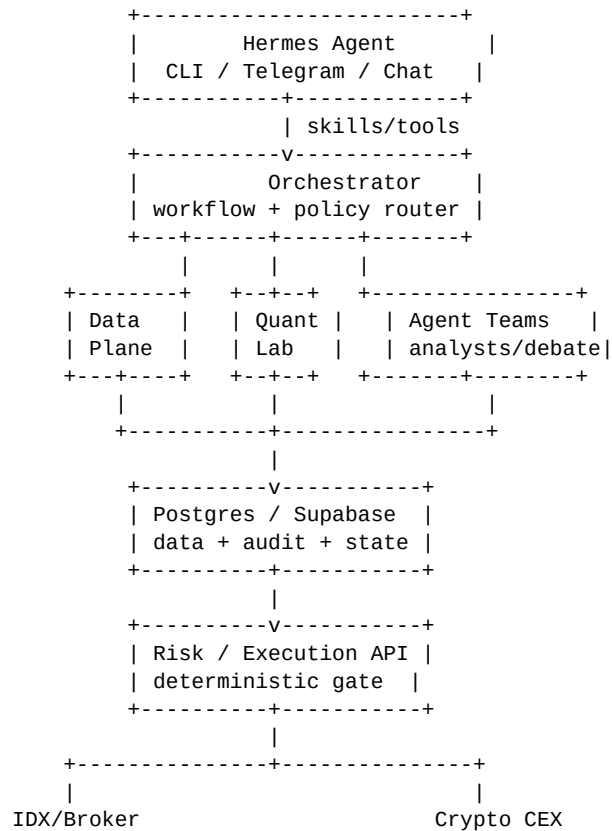
```
{
  "recommendation_id": "rec...",
  "instrument_id": "...",
  "decision": "WATCH|PAPER|LIVE_PROPOSAL|NO_TRADE",
  "direction": "LONG|SHORT|FLAT",
  "timeframe": "...",
  "thesis": "...",
  "entry": {"type": "price|condition", "value": ...},
  "stop": {...},
  "targets": [...],
  "size": {...},
  "invalidation": [...],
  "evidence_ids": [...],
  "risk_check_id": "...",
  "confidence": "LOW|MEDIUM|HIGH",
  "expires_at": "...",
  "status": "AWAITING_REVIEW"
}
```

11.3 Tool/API boundaries

Interface	Allowed	Forbidden
research.read	Fetch evidence and derived analytics	Placing orders
risk.check	Calculate and validate sizing/exposure	Changing its own policy
portfolio.propose	Create recommendation artifact	Sending exchange order
execution.precheck	Validate explicit order request against hard limits	Interpreting free text
execution.submit	Submit approved structured order	Taking an LLM-generated sentence as order input
execution.reconcile	Read order/fill/account state	Silently retrying unknown orders
journal.write	Record outcome and review	Editing raw execution state

12. Deployment architecture

12.1 Recommended architecture



12.2 Laptop-first deployment

Best for initial development and paper trading. Run Hermes + Python services + local Postgres or Supabase local stack in Docker. Persist datasets to a dedicated directory and keep credentials in a local secrets manager / environment file excluded from git. This setup makes debugging, backtests and agent skill development easy.

12.3 Supabase deployment

Recommended cloud backend for because the product is relational and audit-heavy. Supabase provides Postgres, Auth, Storage, Realtime, Edge Functions, scheduled jobs via pg_cron/pg_net, and pgvector. Use RLS on exposed tables and keep privileged service-role credentials server-side. The heavy quant/backtest workers should remain in Python containers; Supabase is the data/control plane, not the primary compute engine for large backtests.

12.4 Firebase deployment

Firebase is viable when the priority is a fast realtime application layer and event-triggered serverless UX. Cloud Functions can run scheduled functions and Firestore-triggered logic. For , however, the relational shape of portfolios, orders, fills, evidence, backtests and audit events makes PostgreSQL/Supabase the simpler system of record. Firebase can still be added later for a mobile-oriented client or push-notification surface.

12.5 Decision matrix

Criterion	Supabase	Firebase	Winner
Relational trading data	Excellent	Good, but document-centric	Supabase

RLS / row authorization	Excellent	paradigm	Supabase
Realtime UI	Strong	Excellent	Tie
Scheduled jobs	Strong via Postgres scheduling	Strong via Cloud Scheduler/Functions	Tie
Quant/Python ecosystem	Easy via external workers	Usually external workers	Supabase
Vector/RAG memory	pgvector	Possible via external services	Supabase
Fast mobile-first UX	Good	Excellent	Firebase
recommendation	PRIMARY	OPTIONAL client/event layer	Supabase

13. UX, reports and daily operati

13.1 Daily cadence

Time (WIB example)	Job
Pre-open / pre-session	Data health, overnight global macro, candidate scan, event calendar
Session start	Market regime update, top movers, liquidity and gap-risk scan
Mid-session	Only if enabled: material news/circuit events and portfolio risk changes
Post-close	End-of-day technical/fundamental refresh, journal updates, backtest queue
Crypto 24/7	Event-driven alerts plus scheduled 4h/12h/24h regime summaries
Weekly	Strategy diagnostics, parameter review, trade-journal behavior review
Monthly	Risk-policy review, data-source audit, model-cost/quality review

13.2 Daily recommendation packet

DAILY DESK - YYYY-MM-DD

1. REGIME
IDX: ... CRYPTO: ...
2. WHAT CHANGED
- ...
3. TOP CANDIDATES
1) ...
2) ...
4. RESEARCH VERDICTS
Bull vs Bear disagreement: ...
5. RISK MAP
Exposure / concentration / event risk / liquidity
6. ACTIONABLE PLANS
WATCH / PAPER / LIVE_PROPOSAL
7. NO-TRADE CONDITIONS
...
8. DATA QUALITY
PASS / WARN / BLOCK
9. VERIFY BEFORE ACTION
...

13.3 User commands through Hermes

Command	Behavior
/ daily	Run both markets daily desk and return ranked candidates.
/ scan IDX	Scan the configured IDX universe.
/ scan CRYPTO	Scan approved crypto universe.
/ research BBCA	Run single-instrument full research workflow.
/ plan BTC/USDT	Generate a research/paper plan, never a live order.
/ backtest <strategy>	Run reproducible backtest and report.
/ shadow	Compare decisions with observed account behavior.
/ risk	Show portfolio risk and active breaches.
/ status	Show system, data, adapter and venue health.
/ kill	Activate system-wide trading kill switch.

13.4 Dashboard

- Daily Desk: regime, candidates, evidence quality, open proposals.
- Portfolio: exposure, drawdown, risk budget, correlation, open positions and orders.

- Research Run: agent tree, evidence, debate, risk m
- Backtest Lab: strategy versions, metrics, walk-forward, stress tests and reproducibility hash.
- Journal: planned vs actual, slippage, adherence, mistakes, lessons.
- System Health: providers, queues, model usage, failed gates, stale data and incident log.

14. Observability, testing and qu

14.1 Observability

Signal	Must capture
Run telemetry	Run ID, parent/child task IDs, agent role, model, tokens, latency, retries
Data telemetry	Source, fetch timestamp, freshness, row counts, checksum, quality verdict
Decision telemetry	Evidence IDs, model outputs, debate turns, risk checks, decision status
Execution telemetry	Order intent, prechecks, venue request/response, fill events, reconciliation
Cost telemetry	Model/API/data spend by workflow, market and strategy
Incident telemetry	Security events, stale data, venue outage, policy breach, manual override

14.2 Test pyramid

- Unit tests for every calculator, rule adapter and schema validator.
- Contract tests for market/exchange/broker adapters.
- Golden-data tests for known instruments and known market events.
- Backtest regression suite covering fees, slippage, corporate actions, funding, partial fills and lookahead.
- Adversarial safety tests: prompt injection, ambiguous symbols, malformed numbers, stale data, contradictory sources, unknown order status.
- End-to-end paper-trading rehearsals before live enablement.

14.3 Release gates

Gate	Required before promotion
G0 Prototype	Schemas, logging, deterministic calculators and one data provider.
G1 Research MVP	Daily desk + research run + backtest + audit ledger.
G2 Paper	Paper engine, journal, reconciliation, health checks, kill switch.
G3 Shadow	Live account read-only integration; compare decisions vs reality.
G4 Guarded Live	Approved venue adapter, secrets isolation, pre-trade gate, small notional, manual approval.
G5 Expanded Live	Only after stable observation period and documented rollback/runbook.

14.4 Safety checklist for every live proposal

1. Instrument identity validated.
2. Price and account data are fresh.
3. Evidence timestamps do not leak future information.
4. Strategy version and risk policy are pinned.
5. Portfolio exposure after trade is within limits.
6. Stop/invalidation exists and is mechanically meaningful.
7. Worst-case notional and liquidity checks pass.
8. Venue rules and order precision pass.
9. No unresolved unknown orders/fills exist.
10. User approval is linked to the exact recommendation ID and constraints.

15. Implementation roadmap

Phase 0 - Foundation (1-2 weeks)

- Create repo, Docker Compose, Python service skeleton, Hermes skill skeleton and typed schemas.
- Implement audit/event model and run IDs.
- Implement provider interface + one crypto exchange adapter + one IDX research data source.
- Implement deterministic risk calculators and system health endpoint.

Phase 1 - Research MVP (2-4 weeks)

- Build daily desk workflow, candidate scanner, technical/fundamental/news agents.
- Add Bull/Bear debate and Research Manager.
- Persist evidence and final recommendation packets.
- Build simple dashboard and Telegram/Hermes reporting.

Phase 2 - Backtest + paper (2-4 weeks)

- Build strategy interface and historical data snapshotting.
- Add fee/slippage models, walk-forward, stress tests and regression suite.
- Build paper portfolio engine and journal/shadow account.
- Add strategy comparison and reproducibility fingerprints.

Phase 3 - Guarded live (3-6 weeks)

- Add one approved broker/venue adapter at a time.
- Add deterministic execution gateway, idempotency and reconciliation.
- Enable small-notional manual-approval live mode only.
- Run incident drills and kill-switch exercises.

Phase 4 - Scale / intelligence

- Portfolio optimizer, cross-market allocation, regime ensemble, alpha library.
- Bounded strategy evolution and model routing based on measured quality.
- Multi-venue execution and richer alternative data.

Definition of Done for MVP

Area	Done when
Research	A single command can produce a full, sourced, auditable recommendation packet for IDX and crypto.
Backtest	Same strategy definition can be tested with reproducible results and leakage checks.
Risk	Every proposal is sized by deterministic risk service and can be vetoed.
Paper	Recommendations become simulated orders with fills and journal entries.
Hermes	Skills can discover/run the workflows with concise operator commands.
Operations	Health, logs, kill switch and incident workflow are tested.

16. Feature comparison and pick-

This section is intended as a decision sheet. " baseline" is the recommended scope in this PRD. Features marked "Adopt" should become part of the initial architecture; "Optional" can be deferred; "Avoid" should not be copied without redesign.

Feature	Source / inspiration	choice	Reason
Structured 5-stage research workflow	Astra reference	Adopt	Foundational operator flow
Human approval before live action	Astra reference	Adopt	Keep as hard control for initial live phases
Specialist analyst team	TradingAgents	Adopt	Fundamental/news/technical/sentiment/macro
Bull/Bear adversarial debate	TradingAgents	Adopt	Reduces confirmation bias; must still allow NO-TRADE
Risk committee with multiple risk voices	TradingAgents	Adopt	Add conservative/neutral/aggressive views
Portfolio manager final decision	TradingAgents	Adopt	Structured status, not direct order call
Director -> Quant -> Risk -> Execution pipeline	AutoHedge	Adopt	Execution as deterministic gateway
Structured JSON outputs	AutoHedge	Adopt	Use typed schemas + versioning
Enterprise-style audit logging	AutoHedge	Adopt	Required for trust and debugging
Swarm intelligence	AutoHedge	Optional	Use selectively; cost can explode
Solana-only autonomous execution	AutoHedge	Avoid	Too narrow for IDX + multi-exchange crypto
Natural-language strategy factory	Moss	Adopt	Excellent operator UX
30+ tunable parameters	Moss	Optional	Expose only curated parameters per strategy
Bit-exact backtest/live parity	Moss	Adopt	Core design rule
Bounded weekly evolution	Moss	P1	Must be out-of-sample validated
Leverage safety guardrails	Moss	Adopt for derivatives	Set 1x default until derivatives gate opens
Multi-market data routing	Vibe-Trading	Adopt	Use provider abstraction
Data-quality / grounding gates	Vibe-Trading	Adopt	Treat as P0
Shadow account	Vibe-Trading	Adopt	Strong bridge from research to live
Trade-journal diagnostics	Vibe-Trading	Adopt	High long-term value
MCP integrations	Vibe-Trading	P1	Useful for tool interoperability
Alpha zoo / factor library	Vibe-Trading	P2	Add after stable data layer
Persistent research memory	Vibe-Trading / Hermes	Adopt	Use with evidence separation
Conversation-driven skills	Hermes	Adopt	Primary operator interface
Progressive disclosure SKILL.md	Hermes	Adopt	Keep skills concise + references on demand
External skill directories	Hermes	P1	Useful for shared team modules
Provider-agnostic LLM routing	TradingAgents	Adopt	Fast/cheap vs deep model tiers
Local-first Docker	TradingAgents / Moss / Vibe	Adopt	Best starting deployment
Supabase as cloud control plane	design	Adopt	Recommended over Firebase for system of record
Firebase as optional client layer	design	Optional	Only if mobile/realtime client needs it
Event-driven alerts	design	Adopt	Thesis/risk/data/venue alerts
Kill switch	design	Adopt	Non-negotiable
Deterministic execution gateway	design	Adopt	LLM never sends raw orders
Order reconciliation / idempotency	design	Adopt before live	Critical for unknown responses
Portfolio correlation / concentration	design	Adopt	Not just single-trade risk
Regime detection	Moss/Vibe patterns	P1	Use as context, not truth
Alternative data / on-chain	design	P1	Only with source quality controls

Automatic production parameter mutation	Generic AI idea	Avoid	rollback	val +
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Recommended feature tiers to pick

Tier	Include
MVP - must have	Daily desk, scanner, analysts, bull/bear, risk, trade plan, data provenance, backtest, paper, journal, kill switch, Hermes skills, Supabase DB
P1 - strong upgrade	Walk-forward, strategy factory, shadow account, macro regime, portfolio optimizer, MCP, bounded evolution, richer on-chain data
P2 - later	Alpha zoo, multi-venue smart routing, autonomous live loop, mobile app, voice, reinforcement learning / model fine-tuning

17. Recommended MBG names

You asked for project names that can naturally shorten to "MBG". These are ranked for clarity and brandability rather than trademark availability (which should be checked separately before publishing).

Name	Expansion	Fit	Notes
Market Brain Guardian	MBG	9/10	Best fit for a research-first agent with strong safety identity.
Multi-Market Brain	MBG	9/10	Signals IDX + crypto without implying execution. Clean, simple.
Market Brain Grid	MBG	8.5/10	Emphasizes multi-agent grid / modular architecture.
Market Balance Guardian	MBG	8/10	Strong risk-management positioning.
Macro Beta Guardian	MBG	7.5/10	More quantitative/institutional feel; less obvious as a general trading desk.
Market Bridge Gateway	MBG	7/10	Good if the product emphasizes execution/data bridges.
Multi-Asset Brain	MBG	7/10	Very clear but "MA" may be more natural abbreviation.

My recommendation: Market Brain Guardian (MBG) as the product name, with " " as the internal architecture / codename. This preserves the Trading Agent working title while giving you an MBG brand direction.

18. Open decisions and build che

Decisions the builder should lock before coding the execution layer

11. Choose first live venue for crypto and first broker/data adapter for IDX.
12. Choose the authoritative production market-data source(s) and fallback policy.
13. Choose default risk budget and portfolio limits.
14. Define initial supported instruments/universe.
15. Choose LLM provider(s) and model tiers.
16. Define retention and privacy policy for research/account data.
17. Confirm local-first vs cloud-first deployment for day-one usage.
18. Approve exact live-mode approval ceremony and emergency policy.

Builder checklist - first 20 implementation tickets

01. Create monorepo with services/ -core, services/data, services/risk, services/execution, ui, hermes-skill.
02. Create Pydantic schemas for Evidence, Run, Recommendation, RiskCheck, OrderIntent, Fill and AuditEvent.
03. Implement Postgres migrations and RLS boundaries.
04. Create ProviderAdapter interface and health contract.
05. Create InstrumentRegistry and market-rule configuration.
06. Build EvidenceLedger service and provenance hashing.
07. Build data-quality gate engine.
08. Implement deterministic indicators and feature engine.
09. Implement scan_candidates workflow.
10. Implement research analysts + typed outputs.
11. Implement Bull/Bear debate + Research Verdict schema.
12. Implement risk calculator + hard policy engine.
13. Implement portfolio proposal service.
14. Implement backtest engine interface + first strategy.
15. Implement paper broker and fill simulator.
16. Implement journal/shadow account.
17. Implement Hermes SKILL.md + CLI facade.
18. Implement daily scheduler and alert dispatcher.
19. Implement dashboard pages: Daily Desk, Research Run, Portfolio, Backtest, Journal, Health.
20. Implement test suite + release gates + kill-switch drill.

Recommended repository structure

```

-trading-agent/
apps/
  dashboard/
  hermes-skill/
services/
  orchestrator/
  data-plane/
  research-agents/
  quant-lab/

```

- risk-engine/
- paper-broker/
- execution-gateway/
- journal/
- packages/
 - schemas/
 - market-rules/
 - strategies/
 - backtest/
 - providers/
- infra/
 - docker/
 - supabase/
- tests/
 - unit/
 - contract/
 - e2e/
 - safety/
- docs/

19. Research sources

Web research was performed on 8 September 2026. Source list below is the research basis for the design decisions in this PRD. The attached reference PDF is also treated as a first-class design input.

A1 - Supplied PDF - AI Trading Bot with ChatGPT Astra: [User-provided attachment; 7 pages. Core workflow: SCAN -> ANALYZE -> RESEARCH -> RISK -> PLAN; human review; no automatic execution.](#)

R1 - NousResearch Hermes Agent - Skills System:

<https://github.com/NousResearch/hermes-agent/blob/main/website/docs/user-guide/features/skills.md>

R2 - NousResearch Hermes Agent - Creating Skills:

<https://github.com/NousResearch/hermes-agent/blob/main/website/docs/developer-guide/creating-skills.md>

R3 - TauricResearch TradingAgents: <https://github.com/tauricresearch/tradingagents>

R4 - TradingAgents paper, arXiv:2412.20138: <https://arxiv.org/abs/2412.20138>

R5 - TradingAgents system architecture reference: <https://deepwiki.com/TauricResearch/TradingAgents/1.1-system-architecture/>

R6 - The Swarm Corporation AutoHedge: <https://github.com/The-Swarm-Corporation/AutoHedge>

R7 - HKUDS Vibe-Trading: <https://github.com/HKUDS/Vibe-Trading>

R8 - Vibe-Trading overview: <https://vibetrading.wiki/docs/latest/getting-started/vibe-trading-overview>

R9 - Moss trade-bot skills: <https://github.com/moss-site/moss-trade-bot-skills>

R10 - Supabase scheduling: <https://supabase.com/docs/guides/functions/schedule-functions>

R11 - Supabase Row Level Security: <https://supabase.com/docs/guides/database/postgres/row-level-security>

R12 - Supabase pgvector: <https://supabase.com/docs/guides/database/extensions/pgvector>

R13 - Firebase scheduled Cloud Functions: <https://firebase.google.com/docs/functions/schedule-functions-1st>

R14 - OJK POJK 27/2024 - Digital Financial Assets incl. Crypto: <https://ojk.go.id/id/regulasi/Pages/POJK-27-2024-AKD-AK.aspx>

R15 - OJK POJK 23/2025 amendment - Digital Financial Assets / Crypto: <https://www.ojk.go.id/id/berita-dan-kegiatan/siaran-pers/Pages/POJK-23-Tahun-2025-Perubahan-POJK-27-Tahun-2024-Penyelenggaraan-Perdagangan-Aset-Kuangan-Digital-Termasuk-Aset-Kripto.aspx>

R16 - OJK PADK 3/2026 implementation rule: <https://ojk.go.id/id/regulasi/Pages/PADK-Nomor-3-Tahun-2026-Penyelenggaraan-Perdagangan-Aset-Kuangan-Digital-termasuk-Aset-Kripto.aspx>

R17 - OJK transition completion notice: <https://ojk.go.id/id/berita-dan-kegiatan/siaran-pers/Pages/Nota-Kesepahaman-OJK-Bappebti-2026.aspx>

Important research / safety notes

Regulatory note: OJK states that responsibility for regulation and supervision of Digital Financial Assets including Crypto transferred from Bappebti to OJK, with the transition formally ended in January 2026. OJK also published POJK 23/2025 and implementation rule PADK 3/2026; the system should therefore keep regulatory assumptions configurable and versioned rather than hard-coded forever.

Engineering note: source projects change rapidly. Re-verify repository status, licenses, APIs, and regulatory requirements at implementation time. The PRD is a build baseline, not a substitute for professional compliance review.

Final recommendation

Build *as a research-first, risk-gated trading operating system - not as a single super-agent that directly trades.*

Use Hermes for the operator experience and workflow skills. Use Python services for data, quant, backtesting, risk, paper execution and deterministic live execution. Use Supabase/Postgres as the persistent control plane. Add Firebase only when a client-facing realtime/mobile surface needs it.

Start with IDX equities + crypto spot, research + paper + shadow modes, strong provenance and audit. Add derivatives and live execution only after the data-quality, backtest, reconciliation and safety test suite is stable. Then add bounded strategy evolution and portfolio intelligence.

The strongest architectural differentiator for is not "more agents". It is trustworthy composition: multiple perspectives, explicit disagreement, deterministic math, evidence provenance, market-specific rules, reproducible research, and a hard boundary between reasoning and money movement.